# Textile Dyes

| Product / Application | BIS | ISO | ASTM | DIN | Other Standards(SSPC,AMPP, REACH, FDA) |
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| Reactive Dye | | ISO 105 Series (Textile Colour Fastness Testing), | ASTM D2056 (Terminology Relating to Textile Colorants); ASTM D629 (Chemical Analysis of Textiles); ASTM D5383 (Visual Evaluation of Colour Difference); ASTM E308 (Spectrophotometric Colour Measurement); ASTM D2244 (Colour Difference Calculation), ASTM G154; ASTM G155 | DIN EN ISO 105 series (Textile Colour Fastness); DIN EN ISO 105-C06 (Wash Fastness); DIN EN ISO 105-X12 (Rubbing Fastness); DIN EN ISO 105-E04 (Perspiration Fastness); DIN EN ISO 105-J01 (Colour Measurement) | REACH: Regulation (EC) No. 1907/2006; REACH Annex XVII (restricted aromatic amines from azo dye impurities); REACH SVHC compliance; FDA: only where used in FDA-regulated food-contact colour applications |
| Direct Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-C06 (Wash Fastness); ISO 105-X12 (Rubbing Fastness), | ASTM D2056 (Textile Colorant Terminology); ASTM D629 (Chemical Analysis); ASTM D5383 (Colour Evaluation); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM G154; ASTM G155 | DIN EN ISO 105 series (Textile Colour Fastness); DIN EN ISO 105-C10 (Washing Fastness); DIN EN ISO 105-X12 (Rubbing Fastness); DIN EN ISO 105-A02 (Grey Scale Evaluation) | REACH: Regulation (EC) No. 1907/2006; REACH Annex XVII (azo compounds and aromatic amines restrictions); FDA: only approved colour additive applications |
| Acid Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-E04 (Perspiration Fastness); ISO 105-X12 (Rubbing Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference); ASTM D5034 (Textile Fabric Strength where applicable), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-E04 (Perspiration Fastness); DIN EN ISO 105-X12 (Rubbing Fastness); DIN EN ISO 105-C06 (Wash Fastness) | REACH: Regulation (EC) No. 1907/2006 (azo dyes, heavy metals, restricted substances); FDA: only where specifically approved as colour additives |
| Basic Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-C06 (Washing Fastness); ISO 105-X12 (Rubbing Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-X12; DIN EN ISO 105-C06; DIN EN ISO 105-J01 (Colour Measurement) | REACH: Regulation (EC) No. 1907/2006 (aromatic amines, heavy metals, restricted substances); FDA: only approved colour additive applications |
| Vat Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-C06 (Washing Fastness); ISO 105-B02 (Light Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-C06; DIN EN ISO 105-X12; DIN EN ISO 105-E04 | REACH: Regulation (EC) No. 1907/2006 (restricted aromatic amines, impurities, SVHC substances); FDA: only approved colour additive applications |
| Sulphur Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-C06 (Washing Fastness); ISO 105-X12 (Rubbing Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-C06; DIN EN ISO 105-X12; DIN EN ISO 105-J01 | REACH: Regulation (EC) No. 1907/2006 (sulphur compounds, restricted substances, heavy metal impurities); FDA: only approved colour additive applications |
| Disperse Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-E04 (Perspiration Fastness); ISO 105-B02 (Light Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-P01 (Dry Heat Fastness); DIN EN ISO 105-C06; DIN EN ISO 105-X12 | REACH: Regulation (EC) No. 1907/2006; REACH Annex XVII (allergenic disperse dyes and restricted aromatic amines); FDA: only approved colour additive applications |
| Azo Dye | | ISO 14362 (Detection of Certain Aromatic Amines from Azo Colourants); ISO 105 Series, | ASTM D629 (Chemical Analysis of Textiles); ASTM D2056 (Colorant Terminology); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM G154; ASTM G155 | DIN EN ISO 14362 series (Detection of Certain Aromatic Amines Derived from Azo Colorants); DIN EN ISO 105 series; DIN EN ISO 105-J01 | REACH: Regulation (EC) No. 1907/2006; REACH Annex XVII Entry 43 (azo dyes releasing restricted aromatic amines); FDA: only approved food/cosmetic colour additive applications |
| Indigo Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-B02 (Light Fastness); ISO 105-X12 (Rubbing Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM G154; ASTM | DIN EN ISO 105 series; DIN EN ISO 105-X12 (Rubbing Fastness); DIN EN ISO 105-C06 (Wash Fastness); DIN EN ISO 105-J01 | REACH: Regulation (EC) No. 1907/2006 (dye impurities, SVHC substances); FDA: only approved colour additive applications |

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| | | | G155 | | |
| Natural Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-B02 (Light Fastness); ISO 105-C06 (Washing Fastness), | ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference); ASTM D5383 (Visual Colour Evaluation), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-J01 (Colour Measurement); DIN EN ISO 105-C06; DIN EN ISO 105-X12 | REACH: Regulation (EC) No. 1907/2006 (natural extracts, contaminants, restricted substances); FDA: applicable only where used as approved food/cosmetic colour additives |
| Synthetic Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-A02/A03 (Colour Assessment); ISO 105-X12 (Rubbing Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-J01; DIN EN ISO 105-C06; DIN EN ISO 105-X12 | REACH: Regulation (EC) No. 1907/2006 (aromatic amines, heavy metals, SVHC substances); FDA: 21 CFR 73/74 series where approved colour additives apply |
| Cationic Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-C06 (Washing Fastness); ISO 105-E04 (Perspiration Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-C06; DIN EN ISO 105-X12; DIN EN ISO 105-E04 | REACH: Regulation (EC) No. 1907/2006 (restricted amines, impurities, environmental hazards); FDA: only approved colour additive applications |
| Anionic Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-C06 (Washing Fastness); ISO 105-X12 (Rubbing Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-C06; DIN EN ISO 105-X12 | REACH: Regulation (EC) No. 1907/2006 (azo compounds, heavy metals, restricted substances); FDA: only approved colour additive applications |
| Pigment Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-X12 (Rubbing Fastness); ISO 105-B02 (Light Fastness), | ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference); ASTM D5383 (Colour Evaluation), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-C06; DIN EN ISO 105-X12; DIN EN ISO 105-A02 | REACH: Regulation (EC) No. 1907/2006 (pigment composition, heavy metals, SVHC substances); FDA: only approved colour additive applications |
| Optical Brightener | | ISO 105-J02 (Instrumental Assessment of Relative Whiteness); ISO 105 Series (Colour Fastness), | ASTM E313 (Whiteness and Yellowness Index); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM D629; ASTM G154; ASTM G155 | DIN EN ISO 105-J01 (Colour Measurement); DIN EN ISO 105-B02 (Colour Fastness to Artificial Light); DIN EN ISO 11664 series | REACH: Regulation (EC) No. 1907/2006 (fluorescent whitening agents, SVHC substances); FDA: applicable only for approved food-contact paper additives |
| Fluorescent Whitening Agent | | ISO 105-J02 (Instrumental Assessment of Whiteness); ISO 105-B02 (Light Fastness), | ASTM E313 (Whiteness Index); ASTM E991 (Color Measurement of Fluorescent Specimens); ASTM E308 (Colour Measurement), ASTM D629; ASTM G154; ASTM G155 | DIN EN ISO 105-J01; DIN EN ISO 105-B02; DIN EN ISO 11664; DIN EN ISO 4892 (UV Stability where applicable) | REACH: Regulation (EC) No. 1907/2006 (optical brighteners, restricted substances); FDA: 21 CFR 176.170 / 176.180 where used in food-contact paper and paperboard applications |
| Textile Printing Dye | | ISO 105 Series (Textile Colour Fastness); ISO 105-X12 (Rubbing Fastness); ISO 105-C06 (Washing Fastness), | ASTM D2056 (Textile Colorants); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference), ASTM G154; ASTM G155 | DIN EN ISO 105 series; DIN EN ISO 105-X12; DIN EN ISO 105-C06; DIN EN ISO 105-E04; DIN EN ISO 11664 | REACH: Regulation (EC) No. 1907/2006; REACH Annex XVII (azo dyes, restricted aromatic amines); FDA: — |
| Leather Dye | | ISO 11640 (Leather Colour Fastness to Rubbing); ISO 11641 (Leather Colour Fastness to Perspiration); ISO 105 Series where applicable, | ASTM D5053 (Colorfastness Evaluation of Leather where applicable); ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement), ASTM G154; ASTM G155 | DIN EN ISO 11640 (Leather Colour Fastness to Rubbing); DIN EN ISO 11642 (Leather Colour Fastness to Water); DIN EN ISO 105 series; DIN EN ISO 11664 | REACH: Regulation (EC) No. 1907/2006; REACH Annex XVII (azo dyes, chromium-related restrictions where applicable); FDA: — |
| Paper Dye | | ISO 5637 (Paper and Board Water Absorption where applicable); ISO 2470 (Paper Brightness); ISO Colour Measurement Standards,ISO 105 | ASTM D589 (Opacity of Paper); ASTM E308 (Colour Measurement); ASTM E313 (Whiteness/Yellowness Index), ASTM D629; ASTM G154; ASTM G155 | DIN EN ISO 5631 series (Paper and Board Colour Measurement); DIN EN ISO 2470 (Brightness); DIN EN ISO 1924 (Tensile Properties where applicable) | REACH: Regulation (EC) No. 1907/2006; FDA: 21 CFR 176.170 and 21 CFR 176.180 where used for food-contact paper/paperboard |

| Product / Application | BIS | ISO | ASTM | DIN | Other Standards(SSPC,AMPP, REACH, FDA) |
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| Food Grade Colour | | ISO 17025 (Testing Requirements where applicable); ISO 1186 Series (Migration Testing for Food Contact Applications); ISO 787 Series (Pigment Testing where applicable),ISO 105 | ASTM D629 (Chemical Analysis); ASTM E308 (Colour Measurement); ASTM E313 (Colour Index/Whiteness where applicable); ASTM D2244 (Colour Difference), ASTM G154; ASTM G155 | DIN EN 1719 (Colouring Agents where applicable); DIN EN 12856 (Food Additives — Spectrophotometric Determination); DIN EN ISO 11664 (Colour Measurement); EU food additive standards adopted as DIN EN | FDA: 21 CFR 70–82 (Colour Additives Regulations); FDA 21 CFR 73/74 series (specific approved colour additives); REACH: Regulation (EC) No. 1907/2006 (SVHC and restricted substances) |

*Source field: Dyes and Pigments*